MATH 116: Probability

Section 8.7

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What Is Probability?

- A number between 0 and 1 describing the likelihood of an event

- To convert to a percentage, multiply by 100
  - Example: "The probability Exam 2 will be on March 23rd is .995"
  - Equivalently: "The probability Exam 2 will be on March 23rd is ...%"

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  - Example: "The probability Exam 2 will be on March 23rd is .995"
  - Equivalently: "The probability Exam 2 will be on March 23rd is 99.5%"
- Need to specify assumptions to calculate probabilities
  - Probabilities are not inherently correct or incorrect, just self-consistent

Example: Rolling a Die

- Assumption 1: The die is fair

Event	Probability
Roll 1	$.1\overline{6}$
Roll 2	$.1\overline{6}$
Roll 3	$.1\overline{6}$
Roll 4	$.1\overline{6}$
Roll 5	$.1\overline{6}$
Roll 6	$.1\overline{6}$

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- Assumption 2: The die is lightest on the side marked 1

Event	Probability
Roll 1	.7
Roll 2	.0625
Roll 3	.0625
Roll 4	.0625
Roll 5	.0625
Roll 6	.05

Example: Rolling a Die

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Roll 1	$.1\overline{6}$
Roll 2	$.1\overline{6}$
Roll 3	$.1\overline{6}$
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Roll 1	.7
Roll 2	.0625
Roll 3	.0625
Roll 4	.0625
Roll 5	.0625
Roll 6	.05

- Question: What is wrong with saying "Rolling each number either happens or it doesn't, so the probability of rolling each number is .5?"

What is a Statistic?

- Statistic: Number that varies randomly
- Examples of statistics:
  - Amount of time (in minutes) someone spends waiting for their bus
  - Height (in feet) of a person selected from the general population
- In this class: Study statistics that have infinitely many possible values, where each has probability 0
  - For instance, the age "27" represents a bunch of exact ages: 27.0, 27.25, 27.5556, 27.99, 27.999,...
- Two equivalent ways to describe a statistic probabilistically:
  - Probability density function (PDF)
  - Cumulative distribution function (CDF)

Probability Density Function (PDF)

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A probability density function for a statistic is a function $p(x)$ such that

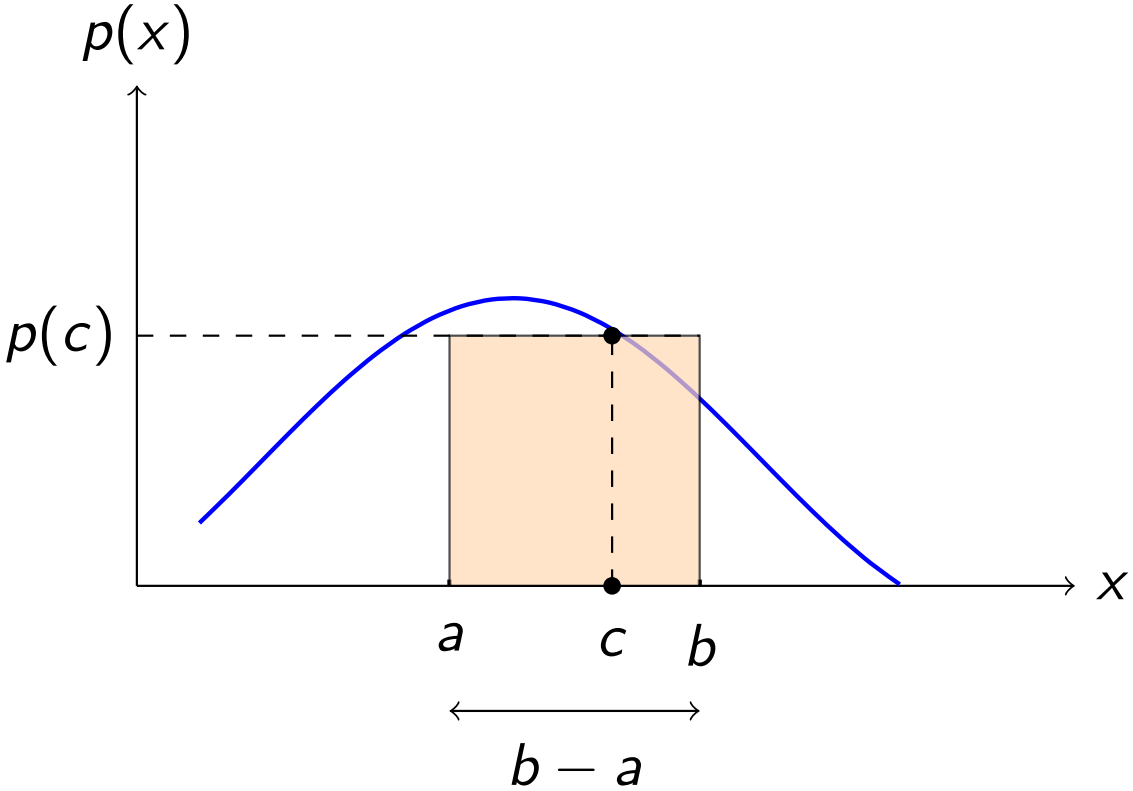
- $p(x) \geq 0$ for all real numbers x; and
- $\int_{-\infty}^{\infty} p(x)dx = 1$

- $p(x)dx =$ infinitely small probability the statistic is equal to x
- Probability that statistic is between a and $b = \int_a^b p(x)dx$
- $\int_{-\infty}^{\infty} p(x)dx = 1$ says that the probability the statistic is equal to something between $-\infty$ and ∞ is 1

Practical Interpretation of PDFs

- If c is in $[a, b]$, then the probability that statistic is between a and b

$$\approx p(c) \cdot (b - a)$$



$$\int_a^b p(x)\,dx \approx p(c)(b - a)$$

Cumulative Distribution Function (CDF)

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A cumulative distribution function for a statistic is a function $P(x)$ such that

- P is increasing: If $a < b$, then $P(a) \leq P(b)$
- $\lim_{x \to -\infty} P(x) = 0$
- $\lim_{x \to \infty} P(x) = 1$

- $P(x) =$ probability the statistic is less than or equal to x
- $\lim_{x \to -\infty} P(x) = 0$ says that the statistic cannot be equal to $-\infty$
- $\lim_{x \to \infty} P(x) = 1$ says that the statistic cannot be equal to ∞

Translating Between PDFs and CDFs

- Suppose a statistic has PDF $p(x)$. How do we obtain its CDF $P(x)$?

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- Summary: The CDF is the antiderivative of the PDF such that $\lim_{x \to -\infty} P(x) = 0$

Example: Selecting a Random Number in [0,6]

- Statistic: Number selected
- Assumption 1: Each number in [0,6] is equally likely
  - PDF: $p(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{6}, & 0 \leq x \leq 6 \\ 0, & x > 6 \end{cases}$

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    - In summary, $P(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{6}, & 0 \leq x \leq 6 \\ 1, & x > 6 \end{cases}$

Example: Waiting Time for a Bus

- Statistic: Waiting time (in minutes)
  - PDF: $p(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{10}, & 0 \leq x \leq 5 \\ \frac{1}{10} e^{-(x-5)/5}, & x > 5 \end{cases}$

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